Electrical Measurements

Voltage is the electrical pressure which forces the current through a circuit.
A **voltmeter** is used to measure voltage.

Look at a voltmeter.
1. Draw the symbol for the voltmeter.

2. What are the units of voltage?

3. What is the largest voltage the voltmeter can be used to measure?

Connect the circuit as shown in the diagram below:

Note: When using a voltmeter remember to connect the black terminal to the negative terminal of the power pack and the red terminal to the positive terminal of the power pack.
Set the voltage of the power pack to 2.
Press the switch and measure the voltage reading.
Increase the voltage of the power pack to 4, then 6, then 8 …. up to 12.
4. What do you notice about the reading on the voltmeter?

5. Will adding another light globe in series, to the circuit, make any difference to the reading on the voltmeter?

Test your prediction.

Resistance and Current

How does the current in a circuit change as the resistance increases?

Hypothesis

Predict what you think is the answer to the question.

Instructions

1. Set up a circuit as shown in the diagram.
2. Ensure that the ammeter is connected to the circuit correctly (ask you teacher to check this).
3. Set the voltage to 12.
4. Press the switch for approximately 15 seconds and measure the current flowing.
5. Measure the current when 2, 3, 4 and 5 globes are connected in series.
6. Record the results in the table below.

Number of globes (resistance)	Current
1	
2	
3	
4	
5	

Questions

1. What happened to the current when the resistance of the circuit was increased?

2. Was your hypothesis correct?

3. What do you predict the current should be if 6 globes are used?

4. Why was it important to keep the voltage constant during the experiment?

5. Why aren’t light globes good resistors to used when you want accurate experimental results?